

To standardize the temperature limits of machine insulation, the National Electrical Manufacturers Association (NEMA) in the United States has defined a series of *insulation system classes*. Each insulation system class specifies the maximum temperature rise permissible for that class of insulation. There are three common NEMA insulation classes for integral-horsepower ac motors: B, F, and H. Each class represents a higher permissible winding temperature than the one before it. For example, the armature winding temperature rise above ambient temperature in one type of continuously operating ac induction motor must be limited to 80°C for class B, 105°C for class F, and 125°C for class H insulation.

The specific temperature specifications for each type of ac motor and generator are set out in great detail in NEMA Standard MG1-1993, *Motors and Generators*. Similar standards have been defined by the International Electrotechnical Commission (IEC) and by various national standards organizations in other countries.

4.7 AC MACHINE POWER FLOWS AND LOSSES

AC generators take in mechanical power and produce electric power, while ac motors take in electric power and produce mechanical power. In either case, not all the power input to the machine appears in useful form at the other end—there is *always* some loss associated with the process.

The efficiency of an ac machine is defined by the equation

$$\eta = \frac{P_{\text{out}}}{P_{\text{in}}} \times 100\% \quad (4-62)$$

The difference between the input power and the output power of a machine is the losses that occur inside it. Therefore,

$$\eta = \frac{P_{\text{in}} - P_{\text{loss}}}{P_{\text{in}}} \times 100\% \quad (4-63)$$

The Losses in AC Machines

The losses that occur in ac machines can be divided into four basic categories:

1. Electrical or copper losses (I^2R losses)
2. Core losses
3. Mechanical losses
4. Stray load losses

ELECTRICAL OR COPPER LOSSES. Copper losses are the resistive heating losses that occur in the stator (armature) and rotor (field) windings of the machine. The stator copper losses (SCL) in a three-phase ac machine are given by the equation

$$P_{\text{SCL}} = 3I_A^2 R_A \quad (4-64)$$

where I_A is the current flowing in each armature phase and R_A is the resistance of each armature phase.

The rotor copper losses (RCL) of a synchronous ac machine (induction machines will be considered separately in Chapter 7) are given by

$$P_{\text{RCL}} = I_F^2 R_F \quad (4-65)$$

where I_F is the current flowing in the field winding on the rotor and R_F is the resistance of the field winding. The resistance used in these calculations is usually the winding resistance at normal operating temperature.

CORE LOSSES. The core losses are the hysteresis losses and eddy current losses occurring in the metal of the motor. These losses were described in Chapter 1. These losses vary as the square of the flux density (B^2) and, for the stator, as the 1.5th power of the speed of rotation of the magnetic fields ($n^{1.5}$).

MECHANICAL LOSSES. The mechanical losses in an ac machine are the losses associated with mechanical effects. There are two basic types of mechanical losses: *friction* and *windage*. Friction losses are losses caused by the friction of the bearings in the machine, while windage losses are caused by the friction between the moving parts of the machine and the air inside the motor's casing. These losses vary as the cube of the speed of rotation of the machine.

The mechanical and core losses of a machine are often lumped together and called the *no-load rotational loss* of the machine. At no load, all the input power must be used to overcome these losses. Therefore, measuring the input power to the stator of an ac machine acting as a motor at no load will give an approximate value for these losses.

STRAY LOSSES (OR MISCELLANEOUS LOSSES). Stray losses are losses that cannot be placed in one of the previous categories. No matter how carefully losses are accounted for, some always escape inclusion in one of the above categories. All such losses are lumped into stray losses. For most machines, stray losses are taken by convention to be 1 percent of full load.

The Power-Flow Diagram

One of the most convenient techniques for accounting for power losses in a machine is the *power-flow diagram*. A power-flow diagram for an ac generator is shown in Figure 4-20a. In this figure, mechanical power is input into the machine, and then the stray losses, mechanical losses, and core losses are subtracted. After they have been subtracted, the remaining power is ideally converted from mechanical to electrical form at the point labeled P_{conv} . The mechanical power that is converted is given by

$$P_{\text{conv}} = \tau_{\text{ind}} \omega_m \quad (4-66)$$

and the same amount of electrical power is produced. However, this is not the power that appears at the machine's terminals. Before the terminals are reached, the electrical $I^2 R$ losses must be subtracted.

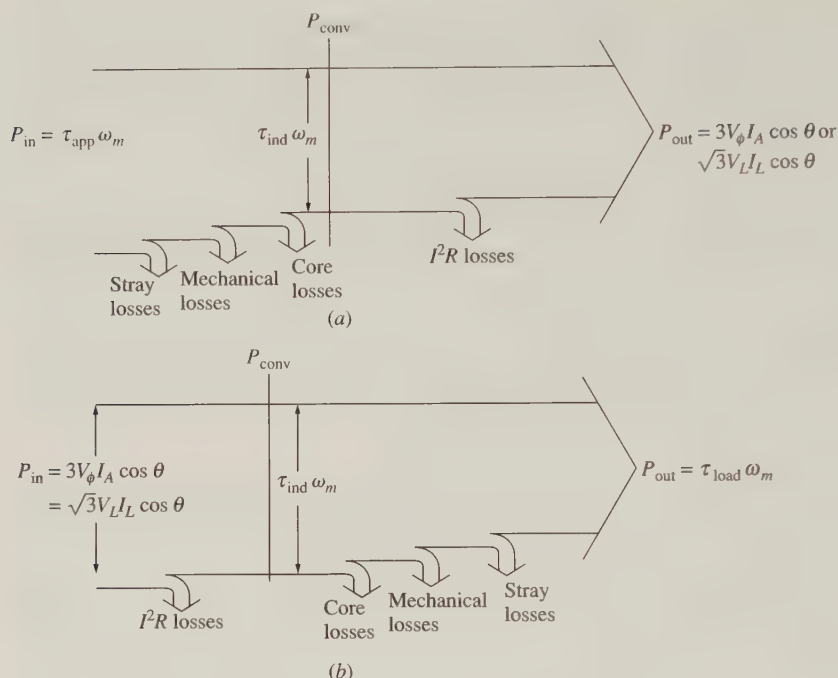


FIGURE 4-20

(a) The power-flow diagram of a three-phase ac generator. (b) The power-flow diagram of a three-phase ac motor.

In the case of ac motors, this power-flow diagram is simply reversed. The power-flow diagram for a motor is shown in Figure 4-20b.

Example problems involving the calculation of ac motor and generator efficiencies will be given in the next three chapters.

4.8 VOLTAGE REGULATION AND SPEED REGULATION

Generators are often compared to each other using a figure of merit called *voltage regulation*. Voltage regulation (VR) is a measure of the ability of a generator to keep a constant voltage at its terminals as load varies. It is defined by the equation

$$VR = \frac{V_{nl} - V_{fl}}{V_{fl}} \times 100\% \quad (4-67)$$

where V_{nl} is the no-load terminal voltage of the generator and V_{fl} is the full-load terminal voltage of the generator. It is a rough measure of the shape of the generator's voltage-current characteristic—a positive voltage regulation means a drooping characteristic, and a negative voltage regulation means a rising characteristic. A small VR is “better” in the sense that the voltage at the terminals of the generator is more constant with variations in load.

Similarly, motors are often compared to each other by using a figure of merit called *speed regulation*. Speed regulation (SR) is a measure of the ability of a motor to keep a constant shaft speed as load varies. It is defined by the equation

$$\text{SR} = \frac{n_{\text{nl}} - n_{\text{fl}}}{n_{\text{fl}}} \times 100\% \quad (4-68)$$

or

$$\text{SR} = \frac{\omega_{\text{nl}} - \omega_{\text{fl}}}{\omega_{\text{fl}}} \times 100\% \quad (4-69)$$

It is a rough measure of the shape of a motor's torque-speed characteristic—a positive speed regulation means that a motor's speed drops with increasing load, and a negative speed regulation means a motor's speed increases with increasing load. The magnitude of the speed regulation tells approximately how steep the slope of the torque-speed curve is.

4.9 SUMMARY

There are two major types of ac machines: synchronous machines and induction machines. The principal difference between the two types is that synchronous machines require a dc field current to be supplied to their rotors, while induction machines have the field current induced in their rotors by transformer action. They will be explored in detail in the next three chapters.

A three-phase system of currents supplied to a system of three coils spaced 120 electrical degrees apart on a stator will produce a uniform rotating magnetic field within the stator. The *direction of rotation* of the magnetic field can be *reversed* by simply swapping the connections to any two of the three phases. Conversely, a rotating magnetic field will produce a three-phase set of voltages within such a set of coils.

In stators of more than two poles, one complete mechanical rotation of the magnetic fields produces more than one complete electrical cycle. For such a stator, one mechanical rotation produces $P/2$ electrical cycles. Therefore, the electrical angle of the voltages and currents in such a machine is related to the mechanical angle of the magnetic fields by

$$\theta_e = \frac{P}{2} \theta_m$$

The relationship between the electrical frequency of the stator and the mechanical rate of rotation of the magnetic fields is

$$f_e = \frac{n_m P}{120}$$

The types of losses that occur in ac machines are electrical or copper losses (I^2R losses), core losses, mechanical losses, and stray losses. Each of these losses was described in this chapter, along with the definition of overall machine efficiency. Finally, voltage regulation was defined for generators as

$$\text{VR} = \frac{V_{\text{nl}} - V_{\text{fl}}}{V_{\text{fl}}} \times 100\%$$

where γ is the angle between $\mathbf{E}_A$ and $\mathbf{I}_A$. The difference between the input power to the generator and the power converted in the generator represents the mechanical, core, and stray losses of the machine.

The real electric output power of the synchronous generator can be expressed in line quantities as

$$P_{\text{out}} = \sqrt{3} V_T I_L \cos \theta \quad (5-16)$$

and in phase quantities as

$$P_{\text{out}} = 3 V_\phi I_A \cos \theta \quad (5-17)$$

The reactive power output can be expressed in line quantities as

$$Q_{\text{out}} = \sqrt{3} V_T I_L \sin \theta \quad (5-18)$$

or in phase quantities as

$$Q_{\text{out}} = 3 V_\phi I_A \sin \theta \quad (5-19)$$

If the armature resistance R_A is ignored (since $X_S \gg R_A$), then a very useful equation can be derived to approximate the output power of the generator. To derive this equation, examine the phasor diagram in Figure 5-16. Figure 5-16 shows a simplified phasor diagram of a generator with the stator resistance ignored. Notice that the vertical segment bc can be expressed as either $E_A \sin \delta$ or $X_S I_A \cos \theta$. Therefore,

$$I_A \cos \theta = \frac{E_A \sin \delta}{X_S}$$

and substituting this expression into Equation (5-17) gives

$$P = \frac{3 V_\phi E_A \sin \delta}{X_S} \quad (5-20)$$

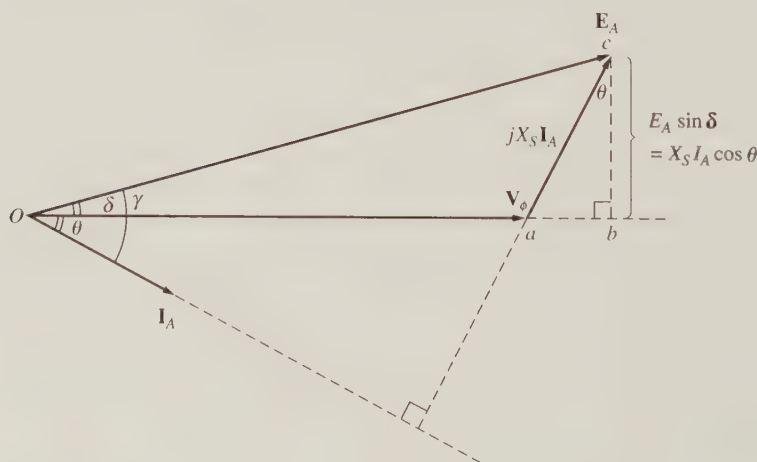


FIGURE 5-16

Simplified phasor diagram with armature resistance ignored.

Since the resistances are assumed to be zero in Equation (5–20), there are no electrical losses in this generator, and this equation is both P_{conv} and P_{out} .

Equation (5–20) shows that the power produced by a synchronous generator depends on the angle δ between $\mathbf{V}_\phi$ and $\mathbf{E}_A$. The angle δ is known as the *torque angle* of the machine. Notice also that the maximum power that the generator can supply occurs when $\delta = 90^\circ$. At $\delta = 90^\circ$, $\sin \delta = 1$, and

$$P_{\text{max}} = \frac{3V_\phi E_A}{X_S} \quad (5-21)$$

The maximum power indicated by this equation is called the *static stability limit* of the generator. Normally, real generators never even come close to that limit. Full-load torque angles of 15 to 20° are more typical of real machines.

Now take another look at Equations (5–17), (5–19), and (5–20). If $\mathbf{V}_\phi$ is assumed constant, then the *real power output is directly proportional* to the quantities $I_A \cos \theta$ and $E_A \sin \delta$, and the reactive power output is directly proportional to the quantity $I_A \sin \theta$. These facts are useful in plotting phasor diagrams of synchronous generators as loads change.

From Chapter 4, the induced torque in this generator can be expressed as

$$\tau_{\text{ind}} = k\mathbf{B}_R \times \mathbf{B}_S \quad (4-58)$$

or as

$$\tau_{\text{ind}} = k\mathbf{B}_R \times \mathbf{B}_{\text{net}} \quad (4-60)$$

The magnitude of Equation (4–60) can be expressed as

$$\tau_{\text{ind}} = kB_R B_{\text{net}} \sin \delta \quad (4-61)$$

where δ is the angle between the rotor and net magnetic fields (the so-called *torque angle*). Since $\mathbf{B}_R$ produces the voltage $\mathbf{E}_A$ and $\mathbf{B}_{\text{net}}$ produces the voltage $\mathbf{V}_\phi$ the angle δ between $\mathbf{E}_A$ and $\mathbf{V}_\phi$ is the same as the angle δ between $\mathbf{B}_R$ and $\mathbf{B}_{\text{net}}$.

An alternative expression for the induced torque in a synchronous generator can be derived from Equation (5–20). Because $P_{\text{conv}} = \tau_{\text{ind}}\omega_m$, the induced torque can be expressed as

$$\tau_{\text{ind}} = \frac{3V_\phi E_A \sin \delta}{\omega_m X_S} \quad (5-22)$$

This expression describes the induced torque in terms of electrical quantities, whereas Equation (4–60) gives the same information in terms of magnetic quantities.

5.7 MEASURING SYNCHRONOUS GENERATOR MODEL PARAMETERS

The equivalent circuit of a synchronous generator that has been derived contains three quantities that must be determined in order to completely describe the behavior of a real synchronous generator:

1. The relationship between field current and flux (and therefore between the field current and E_A)
2. The synchronous reactance
3. The armature resistance

This section describes a simple technique for determining these quantities in a synchronous generator.

The first step in the process is to perform the *open-circuit test* on the generator. To perform this test, the generator is turned at the rated speed, the terminals are disconnected from all loads, and the field current is set to zero. Then the field current is gradually increased in steps, and the terminal voltage is measured at each step along the way. With the terminals open, $I_A = 0$, so E_A is equal to V_ϕ . It is thus possible to construct a plot of E_A or V_T versus I_F from this information. This plot is the so-called *open-circuit characteristic* (OCC) of a generator. With this characteristic, it is possible to find the internal generated voltage of the generator for any given field current. A typical open-circuit characteristic is shown in Figure 5-17a. Notice that at first the curve is almost perfectly linear, until some saturation is observed at high field currents. The unsaturated iron in the frame of the synchronous machine has a reluctance several thousand times lower than the air-gap reluctance, so at first almost *all* the magnetomotive force is across the air gap, and the resulting flux increase is linear. When the iron finally saturates, the reluctance of the iron increases dramatically, and the flux increases much more slowly with an increase in magnetomotive force. The linear portion of an OCC is called the *air-gap line* of the characteristic.

The second step in the process is to conduct the *short-circuit test*. To perform the short-circuit test, adjust the field current to zero again and short-circuit the terminals of the generator through a set of ammeters. Then the armature current I_A or the line current I_L is measured as the field current is increased. Such a plot is called a *short-circuit characteristic* (SCC) and is shown in Figure 5-17b. It is essentially a straight line. To understand why this characteristic is a straight line, look at the equivalent circuit in Figure 5-12 when the terminals of the machine are short-circuited. Such a circuit is shown in Figure 5-18a. Notice that when the terminals are short-circuited, the armature current I_A is given by

$$\mathbf{I}_A = \frac{\mathbf{E}_A}{R_A + jX_S} \quad (5-23)$$

and its magnitude is just given by

$$I_A = \frac{E_A}{\sqrt{R_A^2 + X_S^2}} \quad (5-24)$$

The resulting phasor diagram is shown in Figure 5-18b, and the corresponding magnetic fields are shown in Figure 5-18c. Since $\mathbf{B}_S$ almost cancels $\mathbf{B}_R$, the net magnetic field $\mathbf{B}_{\text{net}}$ is *very* small (corresponding to internal resistive and inductive drops only). Since the net magnetic field in the machine is so small, the machine is unsaturated and the SCC is linear.

To understand what information these two characteristics yield, notice that, with V_ϕ equal to zero in Figure 5-18, the *internal machine impedance* is given by

pens to the motor. Note that an increase in field current increases the magnitude of $\mathbf{E}_A$ but does not affect the real power supplied by the motor. The power supplied by the motor changes only when the shaft load torque changes. Since a change in I_F does not affect the shaft speed n_m , and since the load attached to the shaft is unchanged, the real power supplied is unchanged. Of course, V_T is also constant, since it is kept constant by the power source supplying the motor. The distances proportional to power on the phasor diagram ($E_A \sin \delta$ and $I_A \cos \theta$) must therefore be constant. When the field current is increased, $\mathbf{E}_A$ must increase, but it can only do so by sliding out along the line of constant power. This effect is shown in Figure 6-8b.

Notice that as the value of $\mathbf{E}_A$ increases, the magnitude of the armature current $\mathbf{I}_A$ first decreases and then increases again. At low $\mathbf{E}_A$, the armature current is lagging, and the motor is an inductive load. It is acting like an inductor-resistor combination, consuming reactive power Q . As the field current is increased, the armature current eventually lines up with $\mathbf{V}_\phi$, and the motor looks purely resistive. As the field current is increased further, the armature current becomes leading, and the motor becomes a capacitive load. It is now acting like a capacitor-resistor combination, consuming negative reactive power $-Q$ or, alternatively, supplying reactive power Q to the system.

A plot of I_A versus I_F for a synchronous motor is shown in Figure 6-9. Such a plot is called a *synchronous motor V curve*, for the obvious reason that it is shaped like the letter V. There are several V curves drawn, corresponding to different real power levels.

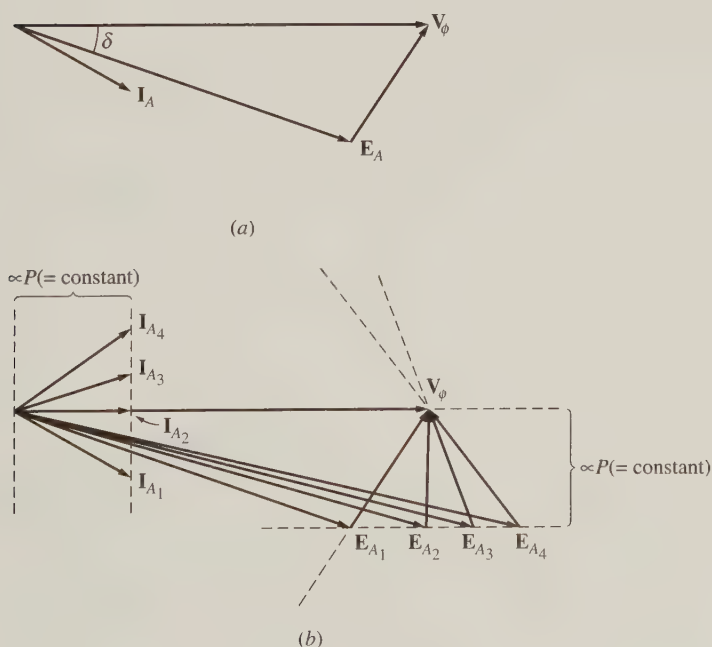


FIGURE 6-8

(a) A synchronous motor operating at a lagging power factor. (b) The effect of an increase in field current on the operation of this motor.

For each curve, the minimum armature current occurs at unity power factor, when only real power is being supplied to the motor. At any other point on the curve, some reactive power is being supplied to or by the motor as well. For field currents *less* than the value giving minimum I_A , the armature current is lagging, consuming Q . For field currents *greater* than the value giving the minimum I_A , the armature current is leading, supplying Q to the power system as a capacitor would. Therefore, by controlling the field current of a synchronous motor, the *reactive power* supplied to or consumed by the power system can be controlled.

When the projection of the phasor E_A onto V_ϕ ($E_A \cos \delta$) is *shorter* than V_ϕ itself, a synchronous motor has a lagging current and consumes Q . Since the field current is small in this situation, the motor is said to be *underexcited*. On the other hand, when the projection of E_A onto V_ϕ is *longer* than V_ϕ itself, a synchronous motor has a leading current and supplies Q to the power system. Since the field current is large in this situation, the motor is said to be *overexcited*. Phasor diagrams illustrating these concepts are shown in Figure 6–10.

Example 6–2. The 208-V, 45-kVA, 0.8-PF-leading, Δ -connected, 60-Hz synchronous motor of the previous example is supplying a 15-hp load with an initial power factor of 0.85 PF lagging. The field current I_F at these conditions is 4.0 A.

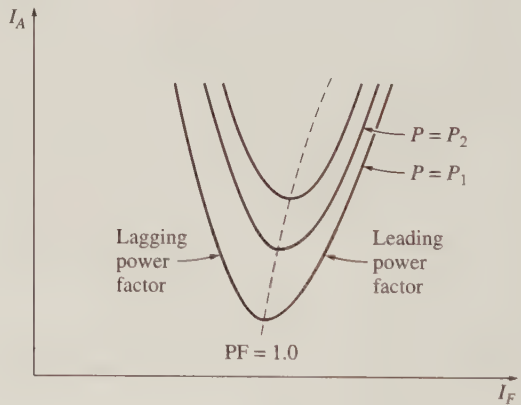


FIGURE 6–9
Synchronous motor V curves.

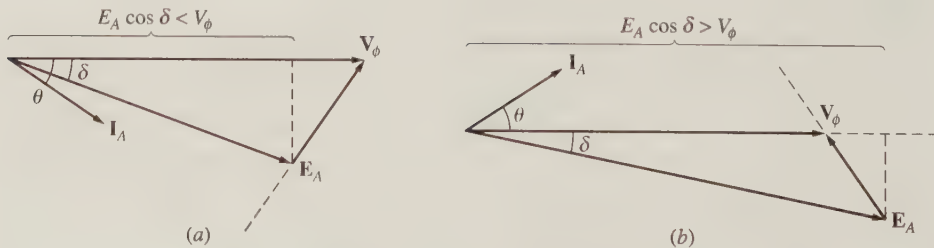


FIGURE 6–10
(a) The phasor diagram of an *underexcited* synchronous motor. (b) The phasor diagram of an *overexcited* synchronous motor.

- Sketch the initial phasor diagram of this motor, and find the values $\mathbf{I}_A$ and $\mathbf{E}_A$.
- If the motor's flux is increased by 25 percent, sketch the new phasor diagram of the motor. What are $\mathbf{E}_A$, $\mathbf{I}_A$, and the power factor of the motor now?
- Assume that the flux in the motor varies linearly with the field current I_F . Make a plot of I_A versus I_F for the synchronous motor with a 15-hp load.

Solution

- From the previous example, the electric input power with all the losses included is $P_{\text{in}} = 13.69 \text{ kW}$. Since the motor's power factor is 0.85 lagging, the resulting armature current flow is

$$\begin{aligned} I_A &= \frac{P_{\text{in}}}{3V_\phi \cos \theta} \\ &= \frac{13.69 \text{ kW}}{3(208 \text{ V})(0.85)} = 25.8 \text{ A} \end{aligned}$$

The angle θ is $\cos^{-1} 0.85 = 31.8^\circ$, so the phasor current $\mathbf{I}_A$ is equal to

$$\mathbf{I}_A = 25.8 \angle -31.8^\circ \text{ A}$$

To find $\mathbf{E}_A$, apply Kirchhoff's voltage law [Equation (6-2)]:

$$\begin{aligned} \mathbf{E}_A &= \mathbf{V}_\phi - jX_S \mathbf{I}_A \\ &= 208 \angle 0^\circ \text{ V} - (j2.5 \Omega)(25.8 \angle -31.8^\circ \text{ A}) \\ &= 208 \angle 0^\circ \text{ V} - 64.5 \angle 58.2^\circ \text{ V} \\ &= 182 \angle -17.5^\circ \text{ V} \end{aligned}$$

The resulting phasor diagram is shown in Figure 6-11, together with the results for part *b*.

- If the flux ϕ is increased by 25 percent, then $E_A = K\phi\omega$ will increase by 25 percent too:

$$E_{A2} = 1.25 E_{A1} = 1.25(182 \text{ V}) = 227.5 \text{ V}$$

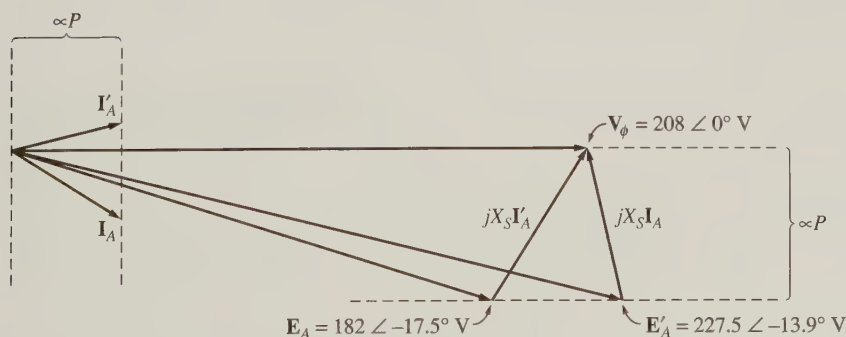
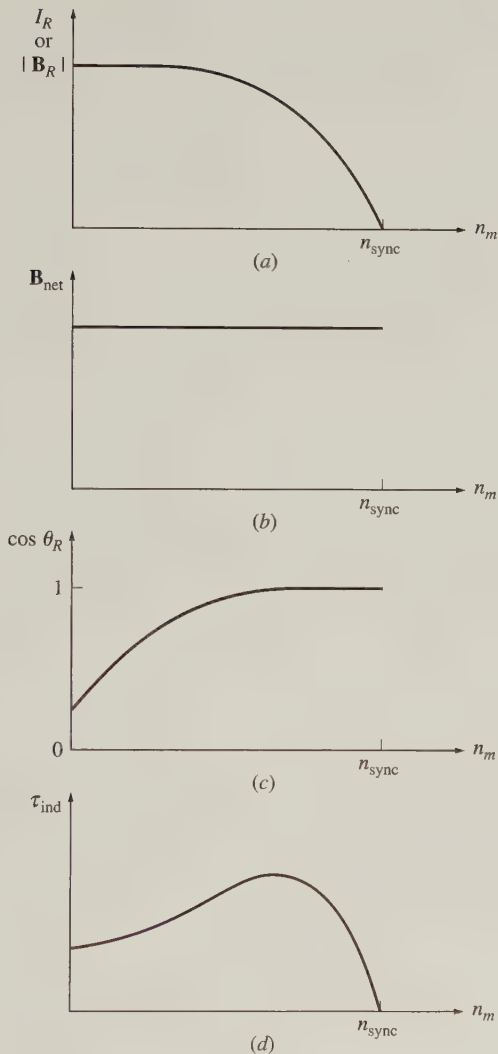


FIGURE 6-11

The phasor diagram of the motor in Example 6-2.

**FIGURE 7-16**

Graphical development of an induction motor torque-speed characteristic. (a) Plot of rotor current (and thus $|\mathbf{B}_R|$) versus speed for an induction motor; (b) plot of net magnetic field versus speed for the motor; (c) plot of rotor power factor versus speed for the motor; (d) the resulting torque-speed characteristic.

decreases approximately linearly with load. In this region of operation, the rotor reactance is negligible, so the rotor power factor is approximately unity, while the rotor current increases linearly with slip. *The entire normal steady-state operating range of an induction motor is included in this linear low-slip region.* Thus in normal operation, an induction motor has a linear speed droop.

The second region on the induction motor's curve can be called the *moderate-slip region*. In the moderate-slip region, the rotor frequency is higher than before, and the rotor reactance is on the same order of magnitude as the rotor resistance. In this region,

the rotor current no longer increases as rapidly as before, and the power factor starts to drop. The peak torque (the *pullout torque*) of the motor occurs at the point where, for an incremental increase in load, the increase in the rotor current is exactly balanced by the decrease in the rotor power factor.

The third region on the induction motor's curve is called the *high-slip region*. In the high-slip region, the induced torque actually decreases with increased load, since the increase in rotor current is completely overshadowed by the decrease in rotor power factor.

For a typical induction motor, the pullout torque on the curve will be 200 to 250 percent of the rated full-load torque of the machine, and the *starting torque* (the torque at zero speed) will be 150 percent or so of the full-load torque. Unlike a synchronous motor, the induction motor can start with a full load attached to its shaft.

The Derivation of the Induction Motor Induced-Torque Equation

It is possible to use the equivalent circuit of an induction motor and the power-flow diagram for the motor to derive a general expression for induced torque as a function of speed. The induced torque in an induction motor is given by Equation (7-35) or (7-36):

$$\tau_{\text{ind}} = \frac{P_{\text{conv}}}{\omega_m} \quad (7-35)$$

$$\tau_{\text{ind}} = \frac{P_{\text{AG}}}{\omega_{\text{sync}}} \quad (7-36)$$

The latter equation is especially useful, since the synchronous speed is a constant for a given frequency and number of poles. Since ω_{sync} is constant, a knowledge of the air-gap power gives the induced torque of the motor.

The air-gap power is the power crossing the gap from the stator circuit to the rotor circuit. It is equal to the power absorbed in the resistance R_2/s . How can this power be found?

Refer to the equivalent circuit given in Figure 7-17. In this figure, the air-gap power supplied to one phase of the motor can be seen to be

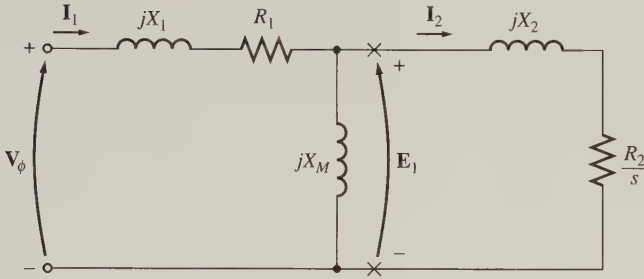
$$P_{\text{AG},1\phi} = I_2^2 \frac{R_2}{s}$$

Therefore, the total air-gap power is

$$P_{\text{AG}} = 3I_2^2 \frac{R_2}{s}$$

If I_2 can be determined, then the air-gap power and the induced torque will be known.

Although there are several ways to solve the circuit in Figure 7-17 for the current I_2 , perhaps the easiest one is to determine the Thevenin equivalent of the portion of the circuit to the left of the X 's in the figure. Thevenin's theorem states that any linear circuit that can be separated by two terminals from the rest of the system can be replaced by a single voltage source in series with an equivalent impedance. If this were done to

**FIGURE 7-17**

Per-phase equivalent circuit of an induction motor.

the induction motor equivalent circuit, the resulting circuit would be a simple series combination of elements as shown in Figure 7-18c.

To calculate the Thevenin equivalent of the input side of the induction motor equivalent circuit, first open-circuit the terminals at the X's and find the resulting open-circuit voltage present there. Then, to find the Thevenin impedance, kill (short-circuit) the phase voltage and find the Z_{eq} seen "looking" into the terminals.

Figure 7-18a shows the open terminals used to find the Thevenin voltage. By the voltage divider rule,

$$\begin{aligned} V_{TH} &= V_{\phi} \frac{Z_M}{Z_M + Z_1} \\ &= V_{\phi} \frac{jX_M}{R_1 + jX_1 + jX_M} \end{aligned}$$

The magnitude of the Thevenin voltage V_{TH} is

$$V_{TH} = V_{\phi} \frac{X_M}{\sqrt{R_1^2 + (X_1 + X_M)^2}} \quad (7-41a)$$

Since the magnetization reactance $X_M \gg X_1$ and $X_M \gg R_1$, the magnitude of the Thevenin voltage is approximately

$$V_{TH} \approx V_{\phi} \frac{X_M}{X_1 + X_M} \quad (7-41b)$$

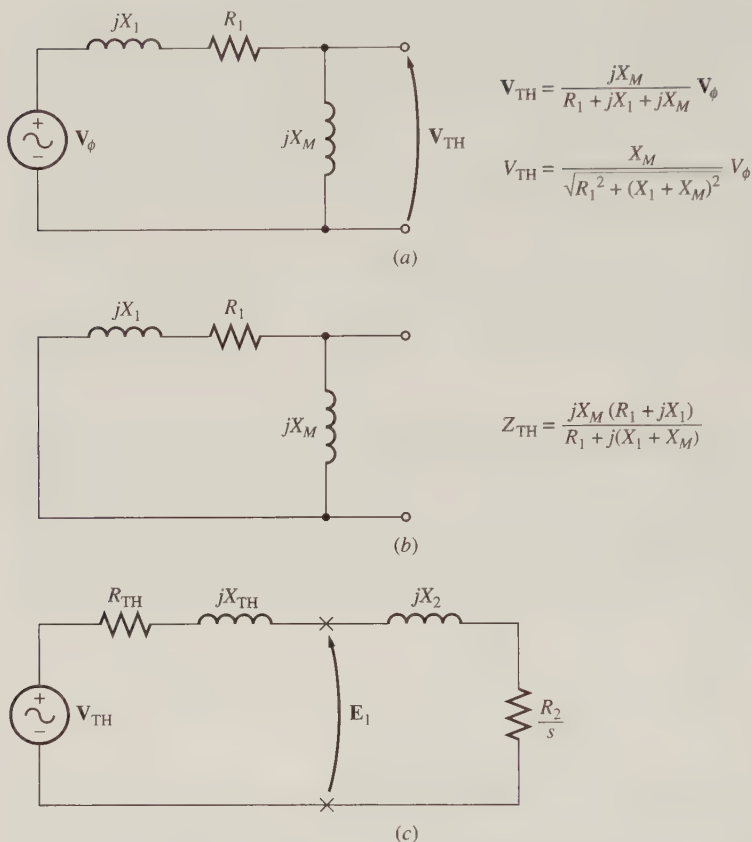
to quite good accuracy.

Figure 7-18b shows the input circuit with the input voltage source killed. The two impedances are in parallel, and the Thevenin impedance is given by

$$Z_{TH} = \frac{Z_1 Z_M}{Z_1 + Z_M} \quad (7-42)$$

This impedance reduces to

$$Z_{TH} = R_{TH} + jX_{TH} = \frac{jX_M(R_1 + jX_1)}{R_1 + j(X_1 + X_M)} \quad (7-43)$$

**FIGURE 7-18**

(a) The Thevenin equivalent voltage of an induction motor input circuit. (b) The Thevenin equivalent impedance of the input circuit. (c) The resulting simplified equivalent circuit of an induction motor.

Because $X_M \gg X_1$ and $X_M + X_1 \gg R_1$, the Thevenin resistance and reactance are approximately given by

$$R_{TH} \approx R_1 \left(\frac{X_M}{X_1 + X_M} \right)^2 \quad (7-44)$$

$$X_{TH} \approx X_1 \quad (7-45)$$

The resulting equivalent circuit is shown in Figure 7-18c. From this circuit, the current I_2 is given by

$$I_2 = \frac{V_{TH}}{Z_{TH} + Z_2} \quad (7-46)$$

$$= \frac{V_{TH}}{R_{TH} + R_2/s + jX_{TH} + jX_2} \quad (7-47)$$

The magnitude of this current is

$$I_2 = \frac{V_{TH}}{\sqrt{(R_{TH} + R_2/s)^2 + (X_{TH} + X_2)^2}} \quad (7-48)$$

The air-gap power is therefore given by

$$\begin{aligned} P_{AG} &= 3I_2^2 \frac{R_2}{s} \\ &= \frac{3V_{TH}^2 R_2/s}{(R_{TH} + R_2/s)^2 + (X_{TH} + X_2)^2} \end{aligned} \quad (7-49)$$

and the rotor-induced torque is given by

$$\tau_{ind} = \frac{P_{AG}}{\omega_{sync}}$$

$$\tau_{ind} = \frac{3V_{TH}^2 R_2/s}{\omega_{sync}[(R_{TH} + R_2/s)^2 + (X_{TH} + X_2)^2]} \quad (7-50)$$

A plot of induction motor torque as a function of speed (and slip) is shown in Figure 7-19, and a plot showing speeds both above and below the normal motor range is shown in Figure 7-20.

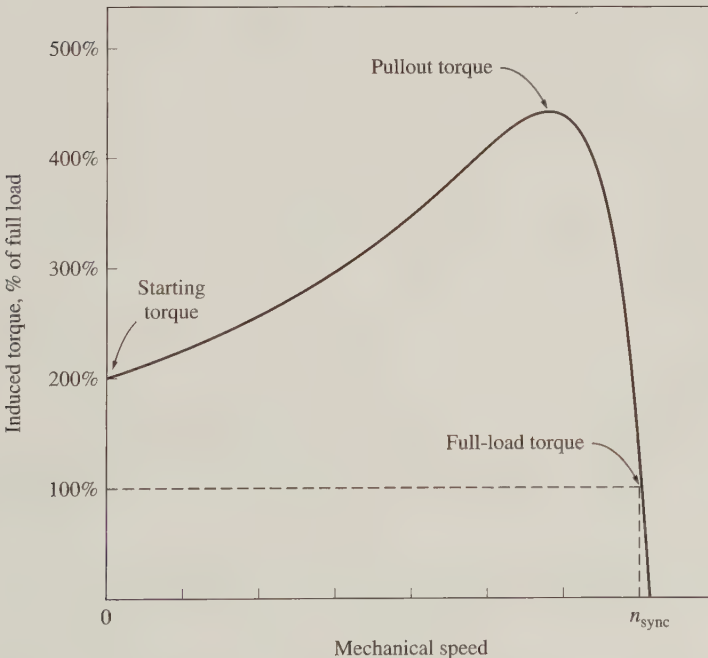
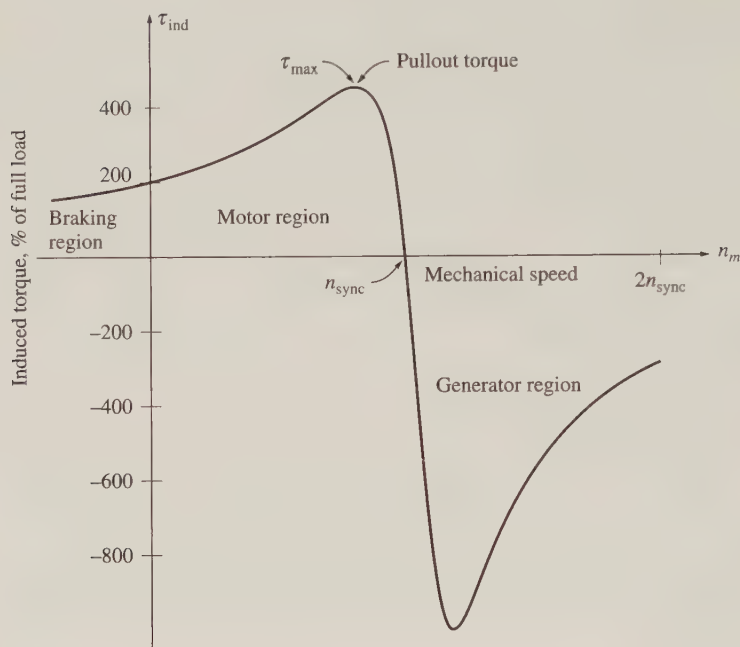


FIGURE 7-19

A typical induction motor torque-speed characteristic curve.

**FIGURE 7-20**

Induction motor torque–speed characteristic curve, showing the extended operating ranges (braking region and generator region).

Comments on the Induction Motor Torque–Speed Curve

The induction motor torque–speed characteristic curve plotted in Figures 7-19 and 7-20 provides several important pieces of information about the operation of induction motors. This information is summarized below:

1. The induced torque of the motor is zero at synchronous speed. This fact has been discussed previously.
2. The torque–speed curve is nearly linear between no load and full load. In this range, the rotor resistance is much larger than the rotor reactance, so the rotor current, the rotor magnetic field, and the induced torque increase linearly with increasing slip.
3. There is a maximum possible torque that cannot be exceeded. This torque, called the *pullout torque* or *breakdown torque*, is 2 to 3 times the rated full-load torque of the motor. The next section of this chapter contains a method for calculating pullout torque.
4. The starting torque on the motor is slightly larger than its full-load torque, so this motor will start carrying any load that it can supply at full power.

5. Notice that the torque on the motor for a given slip varies as the square of the applied voltage. This fact is useful in one form of induction motor speed control that will be described later.
6. If the rotor of the induction motor is driven faster than synchronous speed, then the direction of the induced torque in the machine reverses and the machine becomes a *generator*, converting mechanical power to electric power. The use of induction machines as generators will be described later.
7. If the motor is turning backward relative to the direction of the magnetic fields, the induced torque in the machine will stop the machine very rapidly and will try to rotate it in the other direction. Since reversing the direction of magnetic field rotation is simply a matter of switching any two stator phases, this fact can be used as a way to very rapidly stop an induction motor. The act of switching two phases in order to stop the motor very rapidly is called *plugging*.

The power converted to mechanical form in an induction motor is equal to

$$P_{\text{conv}} = \tau_{\text{ind}} \omega_m$$

and is shown plotted in Figure 7–21. Notice that the peak power supplied by the induction motor occurs at a different speed than the maximum torque; and, of course, no power is converted to mechanical form when the rotor is at zero speed.

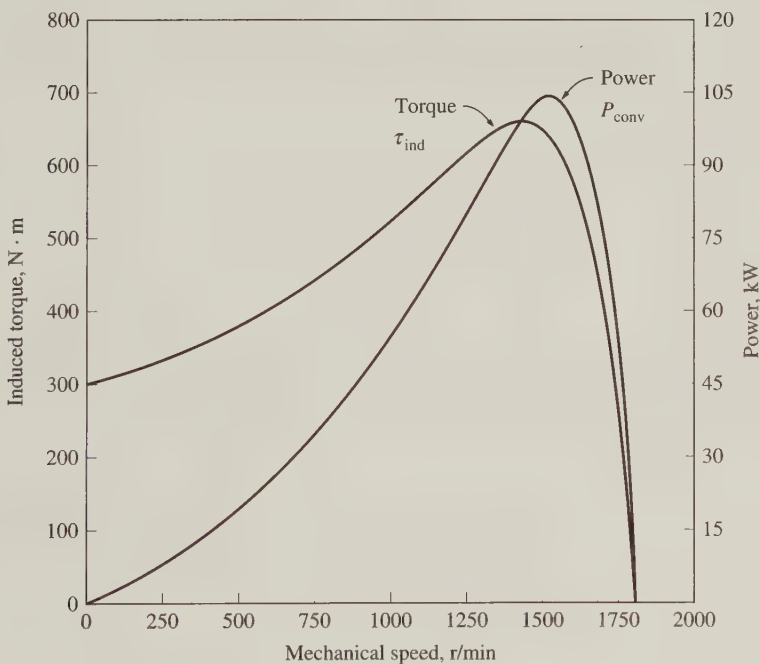


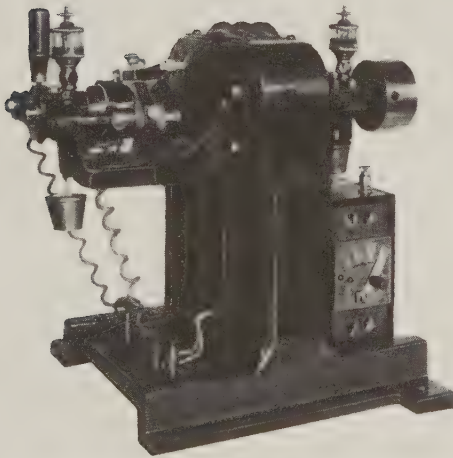
FIGURE 7–21

Induced torque and power converted versus motor speed in revolutions per minute for an example four-pole induction motor.

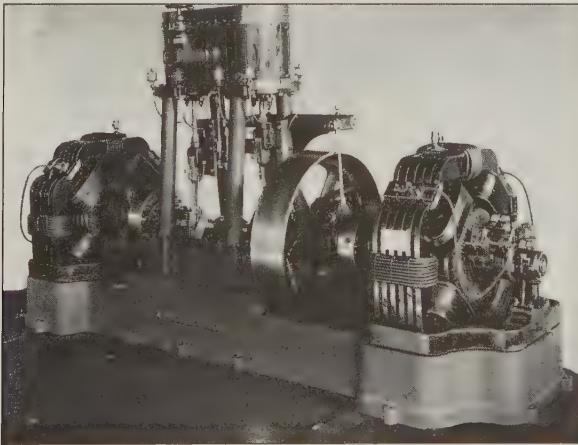
$$\text{SR} = \frac{\omega_{nl} - \omega_{fl}}{\omega_{fl}} \times 100\% \quad (9-1)$$

$$\text{SR} = \frac{n_{nl} - n_{fl}}{n_{fl}} \times 100\% \quad (9-2)$$

It is a rough measure of the shape of a motor's torque-speed characteristic—a positive speed regulation means that a motor's speed drops with increasing load, and a negative speed regulation means a motor's speed increases with increasing load. The magnitude



(a)



(b)

FIGURE 9-1

Early dc motors. (a) A very early dc motor built by Elihu Thompson in 1886. It was rated at about $\frac{1}{2}$ hp. (Courtesy of General Electric Company.) (b) A larger four-pole dc motor from about the turn of the century. Notice the handle for shifting the brushes to the neutral plane. (Courtesy of General Electric Company.)

of the speed regulation tells approximately how steep the slope of the torque-speed curve is.

DC motors are, of course, driven from a dc power supply. Unless otherwise specified, *the input voltage to a dc motor is assumed to be constant*, because that assumption simplifies the analysis of motors and the comparison between different types of motors.

There are five major types of dc motors in general use:

1. The separately excited dc motor
2. The shunt dc motor
3. The permanent-magnet dc motor
4. The series dc motor
5. The compounded dc motor

Each of these types will be examined in turn.

9.2 THE EQUIVALENT CIRCUIT OF A DC MOTOR

The equivalent circuit of a dc motor is shown in Figure 9–2. In this figure, the armature circuit is represented by an ideal voltage source E_A and a resistor R_A . This representation is really the Thevenin equivalent of the entire rotor structure, including rotor coils, interpoles, and compensating windings, if present. The brush voltage drop is represented by a small battery V_{brush} opposing the direction of current flow in the machine. The field coils, which produce the magnetic flux in the generator, are represented by inductor L_F and resistor R_F . The separate resistor R_{adj} represents an external variable resistor used to control the amount of current in the field circuit.

There are a few variations and simplifications of this basic equivalent circuit. The brush drop voltage is often only a very tiny fraction of the generated voltage in a machine. Therefore, in cases where it is not too critical, the brush drop voltage may be left out or approximately included in the value of R_A . Also, the internal resistance of the field coils is sometimes lumped together with the variable resistor, and the total is called R_F (see Figure 9–2*b*). A third variation is that some generators have more than one field coil, all of which will appear on the equivalent circuit.

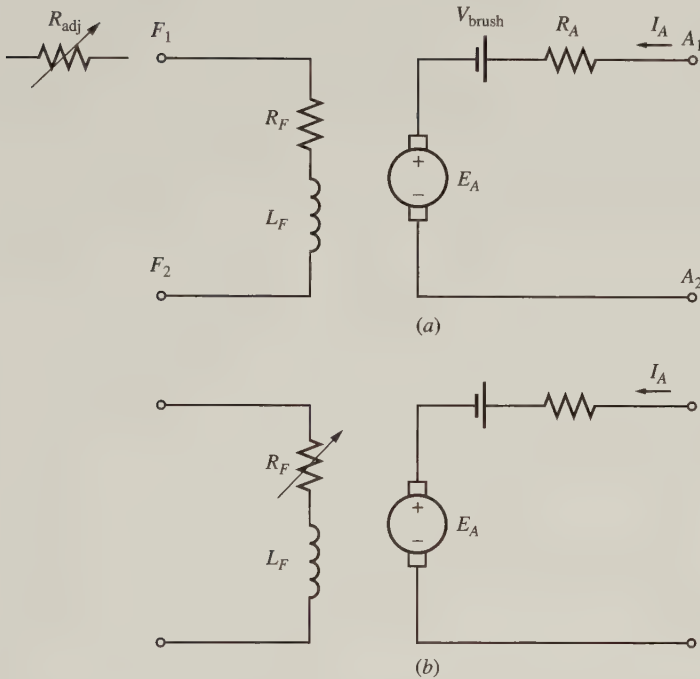
The internal generated voltage in this machine is given by the equation

$$E_A = K\phi\omega \quad (8-38)$$

and the induced torque developed by the machine is given by

$$\tau_{\text{ind}} = K\phi I_A \quad (8-49)$$

These two equations, the Kirchhoff's voltage law equation of the armature circuit and the machine's magnetization curve, are all the tools necessary to analyze the behavior and performance of a dc motor.

**FIGURE 9-2**

(a) The equivalent circuit of a dc motor. (b) A simplified equivalent circuit eliminating the brush voltage drop and combining R_{adj} with the field resistance.

9.3 THE MAGNETIZATION CURVE OF A DC MACHINE

The internal generated voltage E_A of a dc motor or generator is given by Equation (8–38):

$$E_A = K\phi\omega \quad (8-38)$$

Therefore, E_A is directly proportional to the flux in the machine and the speed of rotation of the machine. How is the internal generated voltage related to the field current in the machine?

The field current in a dc machine produces a field magnetomotive force given by $\mathcal{F} = N_F I_F$. This magnetomotive force produces a flux in the machine in accordance with its magnetization curve (Figure 9–3). Since the field current is directly proportional to the magnetomotive force and since E_A is directly proportional to the flux, it is customary to present the magnetization curve as a plot of E_A versus field current for a given speed ω_0 (Figure 9–4).

It is worth noting here that, to get the maximum possible power per pound of weight out of a machine, most motors and generators are designed to operate near the saturation point on the magnetization curve (at the knee of the curve). This implies that

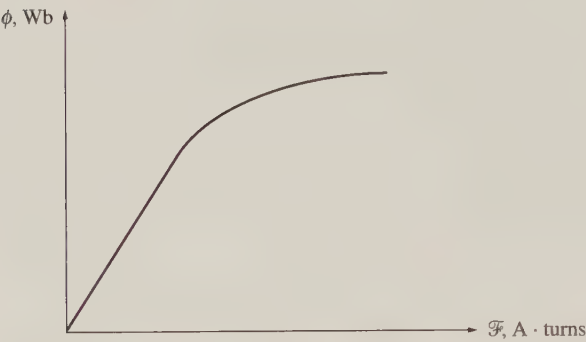


FIGURE 9-3
The magnetization curve of a ferromagnetic material (ϕ versus $\mathcal{F}$).

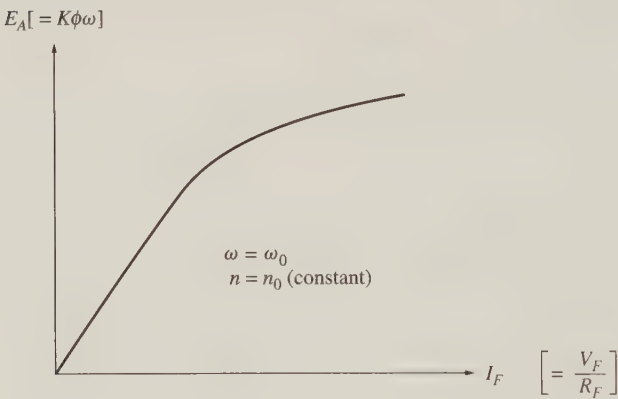


FIGURE 9-4
The magnetization curve of a dc machine expressed as a plot of E_A versus I_F , for a fixed speed ω_o .

a fairly large increase in field current is often necessary to get a small increase in E_A when operation is near full load.

The dc machine magnetization curves used in this book are also available in electronic form to simplify the solution of problems by MATLAB. Each magnetization curve is stored in a separate MAT file. Each MAT file contains three variables: `if_values`, containing the values of the field current; `ea_values`, containing the corresponding values of E_A ; and `n_0`, containing the speed at which the magnetization curve was measured in units of revolutions per minute.

9.4 SEPARATELY EXCITED AND SHUNT DC MOTORS

The equivalent circuit of a separately excited dc motor is shown in Figure 9-5a, and the equivalent circuit of a shunt dc motor is shown in Figure 9-5b. A separately excited dc

We know that the internal generated voltage E_{A0} would be 233 V at a speed of 1200 r/min. Since the actual internal generated voltage E_A is 238.3 V, the actual operating speed of the motor must be

$$\frac{E_A}{E_{A0}} = \frac{n}{n_0} \quad (9-13)$$

$$n = \frac{E_A}{E_{A0}} n_0 = \frac{238.3 \text{ V}}{233 \text{ V}} (1200 \text{ r/min}) = 1227 \text{ r/min}$$

- (b) At 200 A of load in Example 9-1, the motor's speed was $n = 1144$ r/min. In this example, the motor's speed is 1227 r/min. *Notice that the speed of the motor with armature reaction is higher than the speed of the motor with no armature reaction.* This relative increase in speed is due to the flux weakening in the machine with armature reaction.
- (c) To derive the torque-speed characteristic of this motor, we must calculate the torque and speed for many different conditions of load. Unfortunately, the demagnetizing armature reaction magnetomotive force is only given for one condition of load (200 A). Since no additional information is available, we will assume that the strength of $\mathcal{F}_{AR}$ varies linearly with load current.

A MATLAB m-file which automates this calculation and plots the resulting torque-speed characteristic is shown below. It performs the same steps as part *a* to determine the speed for each load current, and then calculates the induced torque at that speed. Note that it reads the magnetization curve from a file called `fig9_9.mat`. This file and the other magnetization curves in this chapter are available for download from the book's World Wide Web site (see Preface for details).

```
% M-file: shunt_ts_curve.m
% M-file create a plot of the torque-speed curve of the
% the shunt dc motor with armature reaction in
% Example 9-2.

% Get the magnetization curve. This file contains the
% three variables if_value, ea_value, and n_0.
load fig9_9.mat

% First, initialize the values needed in this program.
v_t = 250;           % Terminal voltage (V)
r_f = 50;            % Field resistance (ohms)
r_a = 0.06;          % Armature resistance (ohms)
i_l = 10:10:300;     % Line currents (A)
n_f = 1200;          % Number of turns on field
f_ar0 = 840;         % Armature reaction @ 200 A (A-t/m)

% Calculate the armature current for each load.
i_a = i_l - v_t / r_f;

% Now calculate the internal generated voltage for
% each armature current.
e_a = v_t - i_a * r_a;
```

```

% Calculate the armature reaction MMF for each armature
% current.
f_ar = (i_a / 200) * f_ar0;

% Calculate the effective field current.
i_f = v_t / r_f - f_ar / n_f;

% Calculate the resulting internal generated voltage at
% 1200 r/min by interpolating the motor's magnetization
% curve.
e_a0 = interp1(if_values,ea_values,i_f,'spline');

% Calculate the resulting speed from Equation (9-13).
n = ( e_a ./ e_a0 ) * n_0;

% Calculate the induced torque corresponding to each
% speed from Equations (8-55) and (8-56).
t_ind = e_a .* i_a ./ (n * 2 * pi / 60);

% Plot the torque-speed curve
plot(t_ind,n,'Color','k','LineWidth',2.0);
hold on;
xlabel('\tau_{ind} (N-m)','Fontweight','Bold');
ylabel('\itn_{m} \rm\bf(r/min)','Fontweight','Bold');
('\bfShunt DC motor torque-speed characteristic')
axis([ 0 600 1100 1300]);
grid on;
hold off;

```

The resulting torque–speed characteristic is shown in Figure 9–10. Note that for any given load, the speed of the motor with armature reaction is higher than the speed of the motor without armature reaction. ●

Speed Control of Shunt DC Motors

How can the speed of a shunt dc motor be controlled? There are two common methods and one less common method in use. The common methods have already been seen in the simple linear machine in Chapter 1 and the simple rotating loop in Chapter 8. The two common ways in which the speed of a shunt dc machine can be controlled are by

1. Adjusting the field resistance R_F (and thus the field flux)
2. Adjusting the terminal voltage applied to the armature.

The less common method of speed control is by

3. Inserting a resistor in series with the armature circuit.

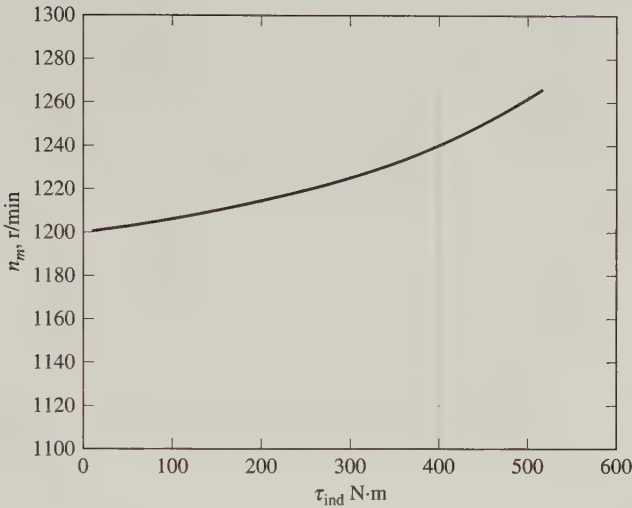


FIGURE 9-10

The torque–speed characteristic of the motor with armature reaction in Example 9-2.

Each of these methods is described in detail below.

CHANGING THE FIELD RESISTANCE. To understand what happens when the field resistor of a dc motor is changed, assume that the field resistor increases and observe the response. If the field resistance increases, then the field current decreases ($I_F = V_T/R_F \uparrow$), and as the field current decreases, the flux ϕ decreases with it. A decrease in flux causes an instantaneous decrease in the internal generated voltage $E_A (= K\phi\omega)$, which causes a large increase in the machine's armature current, since

$$I_A \uparrow = \frac{V_T - E_A \downarrow}{R_A}$$

The induced torque in a motor is given by $\tau_{\text{ind}} = K\phi I_A$. Since the flux ϕ in this machine decreases while the current I_A increases, which way does the induced torque change? The easiest way to answer this question is to look at an example. Figure 9-11 shows a shunt dc motor with an internal resistance of 0.25Ω . It is currently operating with a terminal voltage of 250 V and an internal generated voltage of 245 V. Therefore, the armature current flow is $I_A = (250 \text{ V} - 245 \text{ V})/0.25 \Omega = 20 \text{ A}$. What happens in this motor if there is a 1 percent decrease in flux? If the flux decreases by 1 percent, then E_A must decrease by 1 percent too, because $E_A = K\phi\omega$. Therefore, E_A will drop to

$$E_{A2} = 0.99 E_{A1} = 0.99(245 \text{ V}) = 242.55 \text{ V}$$

The armature current must then rise to

$$I_A = \frac{250 \text{ V} - 242.55 \text{ V}}{0.25 \Omega} = 29.8 \text{ A}$$

Thus a 1 percent decrease in flux produced a 49 percent increase in armature current.